

Neuroergonomic Burden and Shared Attention

Neuroergonomic Burden and Shared Attention I

Neuroergonomics studies brain and behavior in real-world tasks rather than only laboratory tasks [ayaz2019neuroergonomics]. DigiPPPiP needs this framing because the desired practice happens in homes, cafes, studios, hospitals, classrooms, and long-distance relationships. Mobile functional near-infrared spectroscopy (fNIRS) and related field methods make it increasingly plausible to study social and cognitive workload in naturalistic environments [dehais 2020grand;moffat2024mobilefnirs]. The same literature also cautions that naturalism raises measurement risk: motion, lighting, posture, fatigue, device familiarity, and partner relationship history can all change what a signal means.

Neuroergonomic Burden and Shared Attention II

The core design rule is restraint. The original paper-based practice worked partly because the interface was nearly invisible: pen, paper, partner, mark [mikhailova2018pppip]. Digital implementations should add capabilities only where they support the dyad: shared visibility, accessibility, persistence, consentful replay, or place anchoring. Tool palettes, notifications, social feeds, and AI suggestions can easily become technofence if they compete with partner attention, a risk already documented in couple-relationship technology research [mcdaniel2016technofence].

Attention is the scarce resource in DigiPPPiP. The first-principles question for any interface feature is whether it increases partner-directed perception, action, repair, or reflection. A feature that increases expressivity while fragmenting attention is not automatically an improvement. A slower or plainer interface may be neuroergonomically superior when it keeps the dyad inside the shared act rather than inside the tool.

Neuroergonomic Burden and Shared Attention III

[@fig:neuroergonomics_flow] operationalizes three design constraints. The flow surface represents challenge-skill balance, drawing on flow research while remaining a conceptual interface diagnostic rather than an experience sample [@csikszentmihalyi1989flow]. The technoferece curve models the cost of interruptions. The attention simplex asserts that attention must be allocated among the partner's marks, one's own marks, and the shared canvas. These are design diagnostics, not biomarkers. Their purpose is to make interface burden visible before a digital feature is treated as relationally helpful.

Neuroergonomic Burden and Shared Attention IV

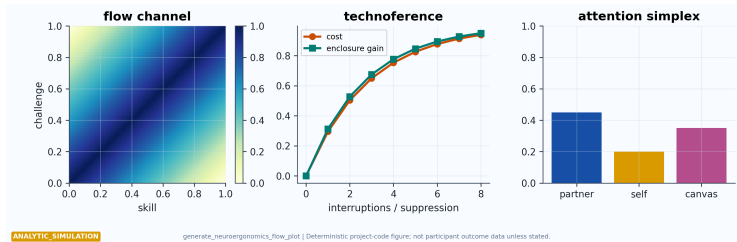


Figure 1: Conceptual neuroergonomic diagnostics: flow channel, technoference cost, intentional-enclosure gain, and attention allocation. The figure is generated by `generate_neuroergonomics_flow_plot()` from `src/neuroergonomics.py`; read the panels as challenge-skill fit, interruption cost versus enclosure gain, and distribution of attention across partner, self, and canvas. It supports interface-burden reasoning. The values are design primitives, not measured workload or flow-state evidence.

Neuroergonomic Burden and Shared Attention V

Brain-computer interface (BCI) and neurofeedback extensions should remain secondary to the human-human practice. Digital art-therapy research suggests that technology can support creative expression and research instrumentation while creating ethical, material, and access constraints [Zubala2021digitalarttherapy; Reitere2024telehealth], but DigiPPPiP should not make neural instrumentation a prerequisite. The strongest design path is progressive enhancement: the practice works with pen and paper, improves with a shared digital canvas, and may later support optional biofeedback for research contexts.